F3XP M.A.R.S.

A Level Physics Practice Paper 3

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

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PHYSICS **9702/43**

Paper 4 A Level Structured Questions **August/September 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 100.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **24** pages. Any blank pages are indicated.

Data

acceleration of free fall	$g = 9.81\ \text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\ \text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\ \text{C}$
unified atomic mass unit	$1\ \text{u} = 1.66 \times 10^{-27}\ \text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\ \text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\ \text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\ \text{mol}^{-1}$
molar gas constant	$R = 8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\ \text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ \text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\ \text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\ \text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$

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gravitational potential	$\phi = -\frac{GM}{r}$
gravitational potential energy	$E_P = -\frac{GMm}{r}$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
simple harmonic motion	$a = -\omega^2 x$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t$ $v = \pm \omega \sqrt{(x_0^2 - x^2)}$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
electrical potential energy	$E_P = \frac{Qq}{4\pi\varepsilon_0 r}$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
discharge of a capacitor	$x = x_0 e^{-\frac{t}{RC}}$
Hall voltage	$V_H = \frac{BI}{ntq}$
alternating current/voltage	$x = x_0 \sin \omega t$
radioactive decay	$x = x_0 e^{-\lambda t}$
decay constant	$\lambda = \frac{0.693}{t_{½}}$
intensity reflection coefficient	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
Doppler redshift	$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$

1 A satellite of mass m moves in a circular orbit of radius r about a planet of mass M, as shown in Fig. 1.1. The planet may be considered to be a point mass at its centre. The gravitational constant is G.

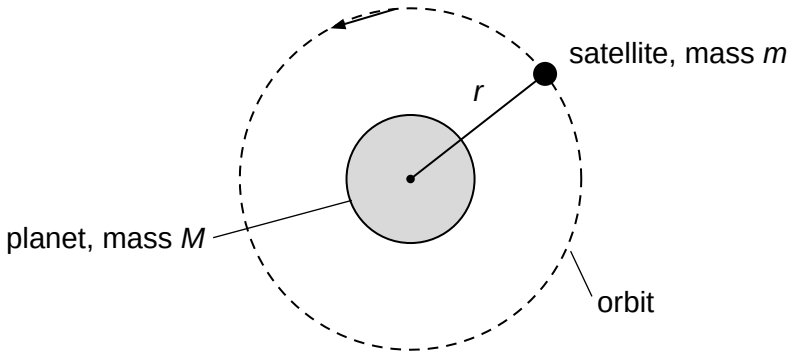


Fig. 1.1 (not to scale)

(a) Show that the period T of the orbit is given by

$$T^2 = \frac{4\pi^2}{GM} r^3.$$

[2]

(b) The Earth has mass 5.98×10^{24} kg and rotates once on its axis every 24.0 hours.

(i) Calculate the radius of a geostationary orbit about the Earth.

radius = ... m [2]

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(ii) A satellite X moves in a circular orbit that has the radius calculated in (b)(i). X moves around the Earth in the same direction as the Earth rotates. The plane of the orbit of X is at an angle of 30° to the plane of the equator, as shown in Fig. 1.2.

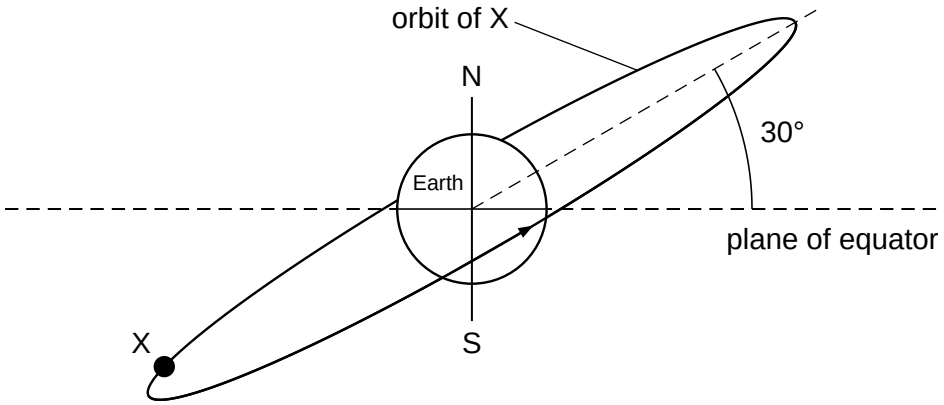


Fig. 1.2 (not to scale)

A student states that X is a geostationary satellite because its orbital period is 24.0 hours. Explain why the student is wrong.

.....

.....

..... [2]

(c) A satellite of mass 1200 kg is moved from a circular orbit of radius 6.80×10^6 m about the Earth to a geostationary orbit.

(i) Determine the minimum energy that must be supplied to the satellite by its rocket engines.

energy = ... J [3]

(ii) The speed of the satellite in the geostationary orbit is less than its speed in the original orbit. Explain how this is possible when energy has been supplied to the satellite.

.....

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..... [2]

[Total: 11]

2 **(a)** State **two** of the basic assumptions of the kinetic theory of gases.

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..... [2]

(b) A constant-volume gas thermometer is shown in Fig. 2.1. The bulb contains a fixed mass of a gas that may be assumed to be ideal.

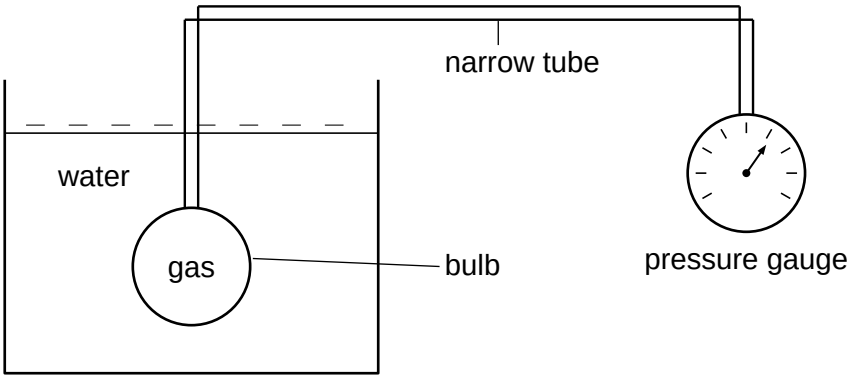


Fig. 2.1

The volume of the gas is kept constant. The bulb is placed in water at different Celsius temperatures θ and the pressure *p* of the gas is measured. The results are shown in Fig. 2.2.

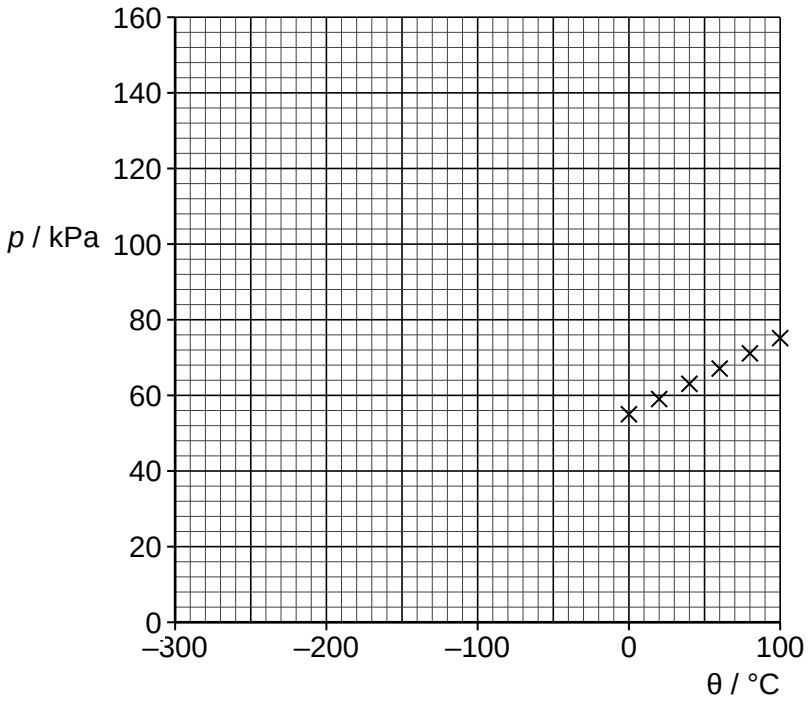


Fig. 2.2

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(i) On Fig. 2.2, draw the best-fit line for the points and extend it to determine the Celsius temperature at which the pressure of the gas would be zero.

temperature = ... °C [2]

(ii) The bulb is emptied and then filled with twice the number of molecules of the same gas. The volume of the bulb is unchanged.

On Fig. 2.2, draw a line to show the variation with θ of *p* for this gas for temperatures from –273 °C to 100 °C. [2]

(c) A thermistor thermometer is calibrated by assuming that the resistance of the thermistor varies linearly with temperature between the ice point (0 °C) and the steam point (100 °C). The resistance of the thermistor is 32.0 kΩ at the ice point and 0.770 kΩ at the steam point.

The thermistor and the bulb of the gas thermometer in (b) are placed together in a liquid. The gas thermometer shows that the temperature of the liquid is 50 °C. The resistance of the thermistor is 3.73 kΩ.

(i) Calculate the temperature of the liquid shown by the thermistor thermometer.

temperature = ... °C [2]

(ii) Both thermometers read correctly at the ice point and at the steam point. Explain why they give different readings for the temperature of the liquid, and state which reading is the thermodynamic temperature of the liquid.

.....

.....

..... [2]

[Total: 10]

3 **(a)** State what is meant by the *internal energy* of a system.

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..... [2]

(b) A fixed mass of an ideal gas is taken through the cycle of changes A → B → C → A shown in Fig. 3.1.

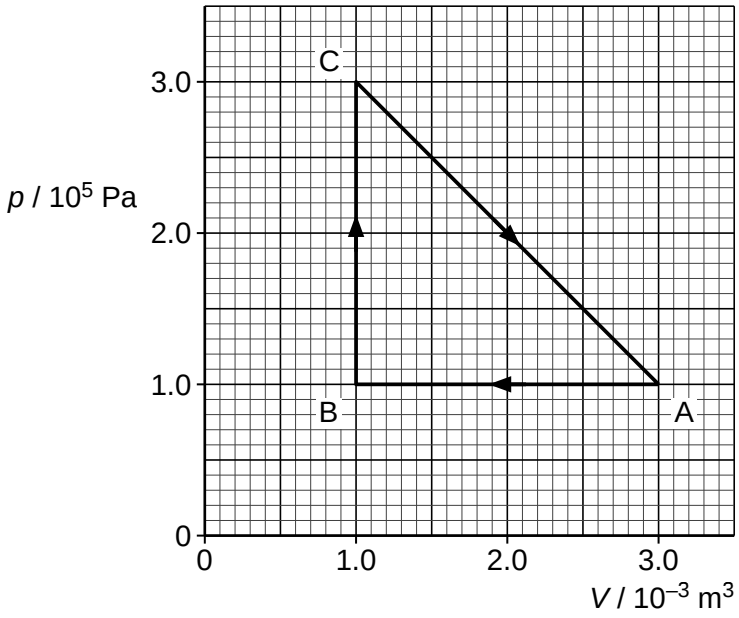


Fig. 3.1

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During A → B, the gas is compressed at constant pressure. During B → C, the gas is heated at constant volume. During C → A, the gas expands and the variation of pressure p with volume V is a straight line.

Table 3.1 shows some of the energy changes during the cycle. W is the work done **on** the gas, q is the thermal energy supplied **to** the gas and ΔU is the increase in internal energy of the gas.

Complete Table 3.1.

Table 3.1

change	W / J	q / J	ΔU / J
A → B	+200	–500	
B → C		+300	
C → A			

[3]

(c) State and explain whether the temperature of the gas is constant during the change C → A.

.....

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..... [2]

(d) A rigid container is surrounded by thermal insulation. The container is divided into two equal sections by a partition, as shown in Fig. 3.2.

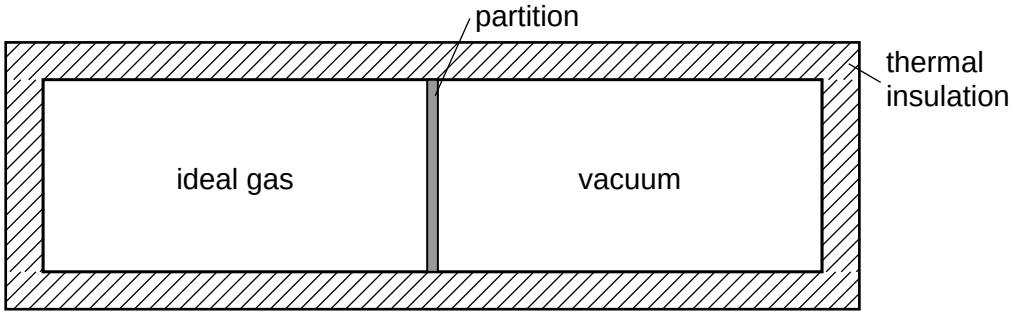


Fig. 3.2

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One section contains an ideal gas and the other section is a vacuum. The partition is removed and the gas expands to fill the whole container.

State and explain whether the temperature of the gas changes.

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..... [3]

[Total: 10]

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4 **(a)** State what is meant by *simple harmonic motion*.

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..... [2]

(b) A glider of mass 0.50 kg on a horizontal air track is attached to a fixed support by a light spring, as shown in Fig. 4.1.

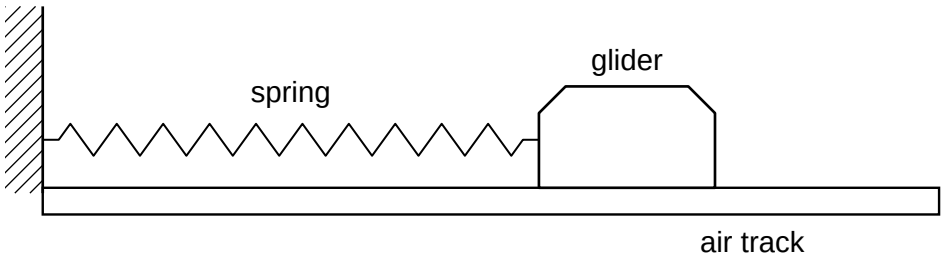


Fig. 4.1

Frictional forces are negligible. The glider is displaced 4.0 cm from its equilibrium position and released. It then oscillates with simple harmonic motion.

Fig. 4.2 shows the variation with displacement x of the elastic potential energy E_P stored in the spring.

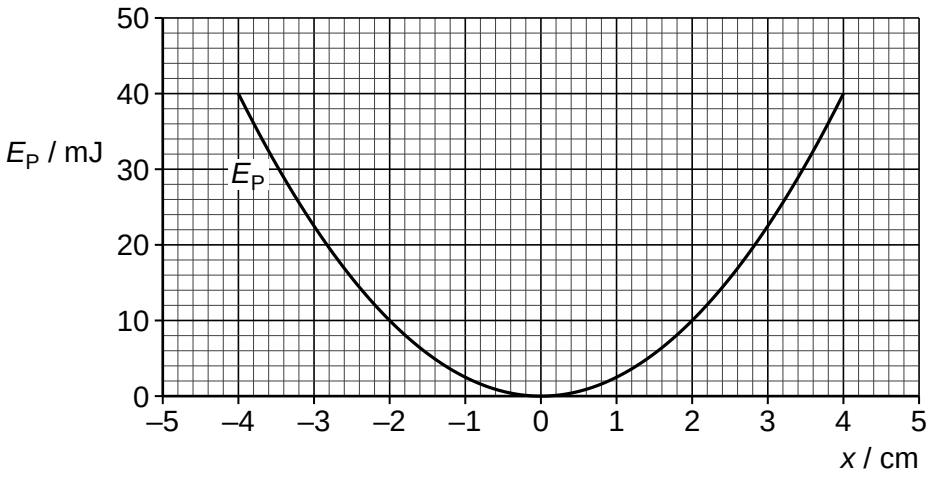


Fig. 4.2

(i) Use Fig. 4.2 to show that the spring constant of the spring is 50 N m^{-1}.

[1]

(ii) On Fig. 4.2, draw lines to show the variation with x of:

1. the kinetic energy of the glider (label this line K)
2. the total energy of the oscillations (label this line T). [2]

(c) The spring is replaced by two springs connected end to end, as shown in Fig. 4.3. Each spring is identical to the original spring.

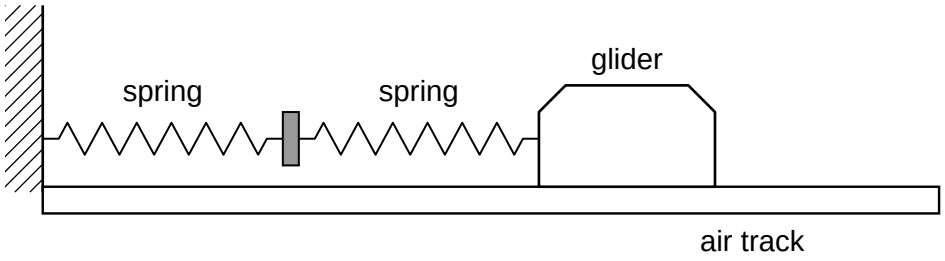


Fig. 4.3

The glider is displaced 2.0 cm from its new equilibrium position and released.

Determine the period of the oscillations of the glider.

period = ... s [3]

(d) The glider is attached again to the single spring of Fig. 4.1, displaced 4.0 cm and released. The air supply to the track is reduced so that the oscillations are lightly damped. After several oscillations, the amplitude has decreased to 2.0 cm.

Determine the energy that has been dissipated.

energy = ... mJ [2]

[Total: 10]

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5 An isolated metal sphere S is held above a large horizontal metal plate that is earthed. The sphere is charged so that its electric potential is +1000 V. Fig. 5.1 shows the lines of equal electric potential (equipotential lines) for potentials of +750 V, +500 V and +250 V around the sphere.

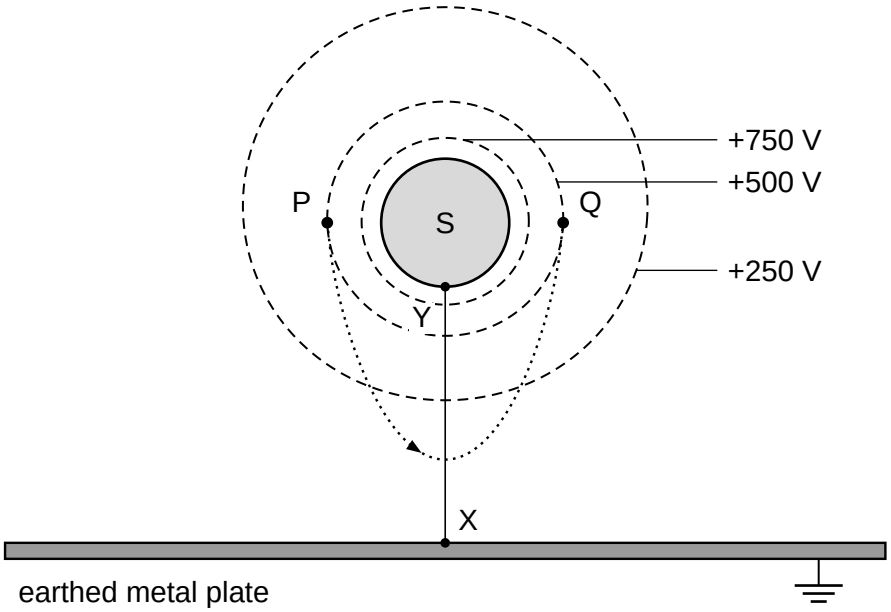


Fig. 5.1

(a) (i) On Fig. 5.1, draw an arrow at point P to show the direction of the electric field at P. [1]

(ii) Points P and Q lie on the same equipotential line. A small positive charge is moved from P to Q along the dotted path shown in Fig. 5.1. State and explain the total work done on the charge by the electric field.

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..... [2]

(b) Point X is on the plate and point Y is on the sphere, vertically above X. Fig. 5.2 shows the variation of the electric potential *V* with distance *x* from X along XY.

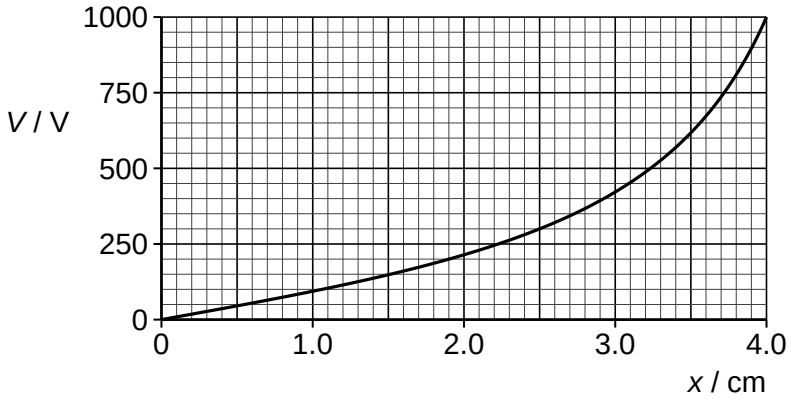


Fig. 5.2

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(i) A student states that the electric field strength at X is zero because the electric potential at X is zero. Use Fig. 5.2 to explain why the student is wrong.

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.....

..... [2]

(ii) A charged oil droplet of mass 7.2×10^{-16} kg is held stationary at the point on XY where $x = 2.0$ cm. Air resistance and upthrust are negligible.

Use Fig. 5.2 to determine the number of excess electrons on the droplet.

number = ... [3]

(iii) The charge on the sphere is now doubled. The droplet is initially at rest at the same point. Determine the magnitude and the direction of the initial acceleration of the droplet.

magnitude = ... m s^{-2}

direction = ... [2]

[Total: 10]

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6 **(a)** Define the *capacitance* of a parallel plate capacitor.

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..... [1]

(b) A parallel plate capacitor is connected to a battery of electromotive force (e.m.f.) 12 V and a switch S, as shown in Fig. 6.1.

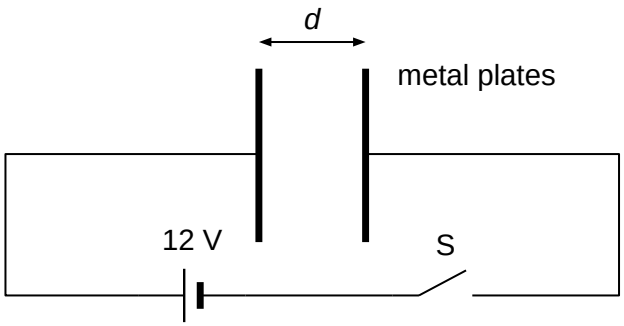


Fig. 6.1

The capacitance of the capacitor is inversely proportional to the separation of its plates. The plates are initially a distance *d* apart and the capacitor is fully charged. Two separate experiments are carried out, each starting from this initial state.

Experiment 1: switch S remains closed while the plates are slowly pulled apart until their separation is 2*d*.
Experiment 2: switch S is opened and then the plates are slowly pulled apart until their separation is 2*d*.

(i) Complete Table 6.1 by giving the ratio *final value* / *initial value* of each quantity. The first row has been completed for you.

Table 6.1

quantity	experiment 1 (S closed)	experiment 2 (S open)
capacitance	0.5	0.5
magnitude of charge on each plate		
potential difference between plates		
energy stored		

[3]

(ii) Work is done to pull the plates apart in both experiments. Explain why, in experiment 1, the energy stored in the capacitor decreases.

.....

.....

..... [2]

(c) A capacitor of capacitance 470 μF is charged to a potential difference (p.d.) of 10.0 V. It is then discharged through a resistor of resistance R. Fig. 6.2 shows the variation with time t of ln (V / V), where V is the p.d. across the capacitor.

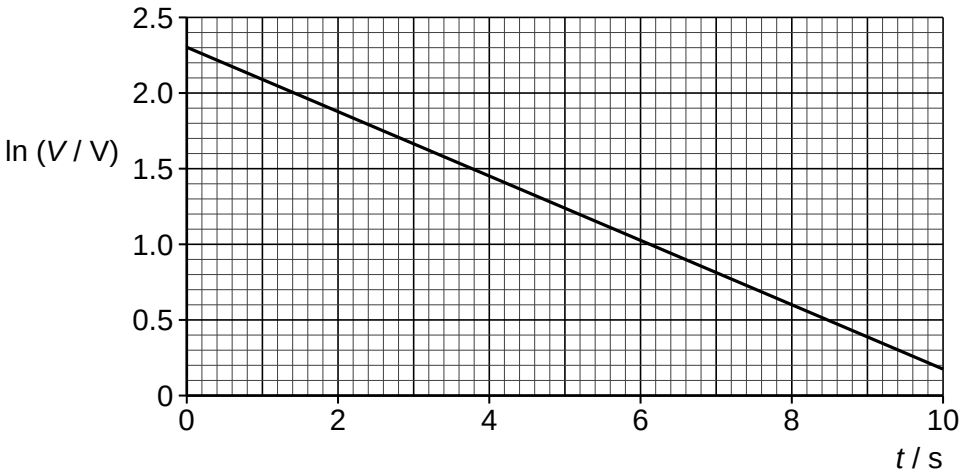


Fig. 6.2

(i) Use Fig. 6.2 to determine R.

R = ... Ω [2]

(ii) Determine the time taken for the energy stored in the capacitor to decrease to half of its initial value.

time = ... s [2]

[Total: 10]

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7 **(a)** State Faraday's law of electromagnetic induction.

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(b) A bar magnet is held vertically above a coil of wire, as shown in Fig. 7.1. The coil is connected to a resistor.

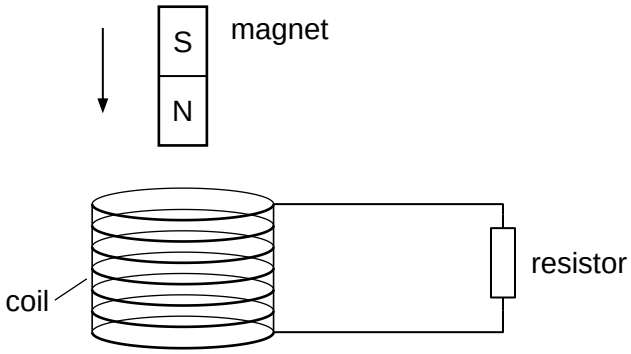


Fig. 7.1

The magnet is released from rest at time $t = 0$ and falls through the coil. Fig. 7.2 shows the variation with t of the magnetic flux linkage $N\Phi$ of the coil.

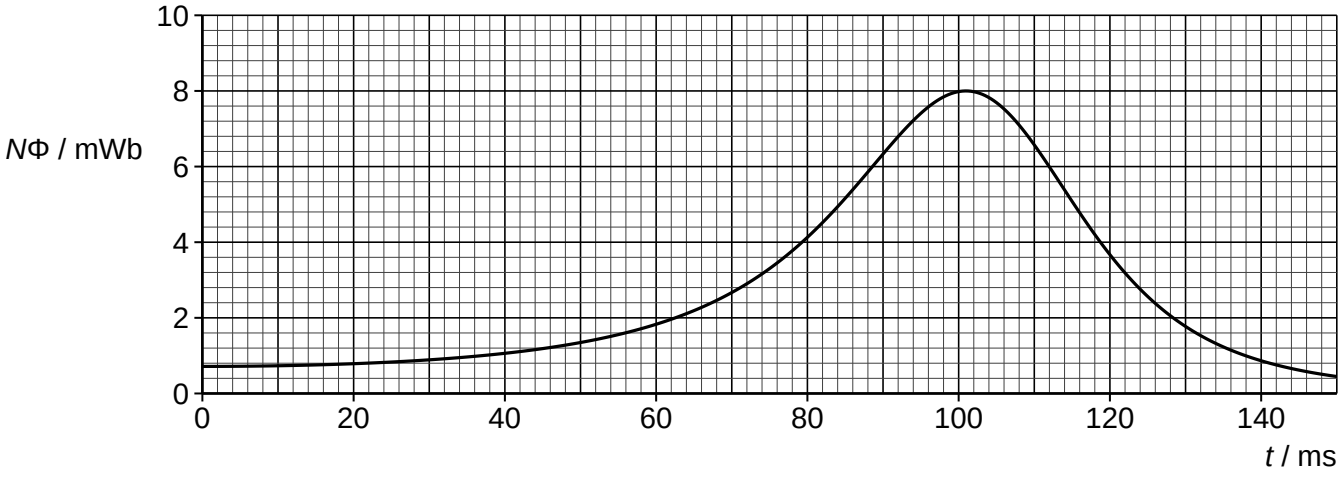


Fig. 7.2

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(i) On Fig. 7.3, sketch the variation with t of the electromotive force (e.m.f.) E induced in the coil.

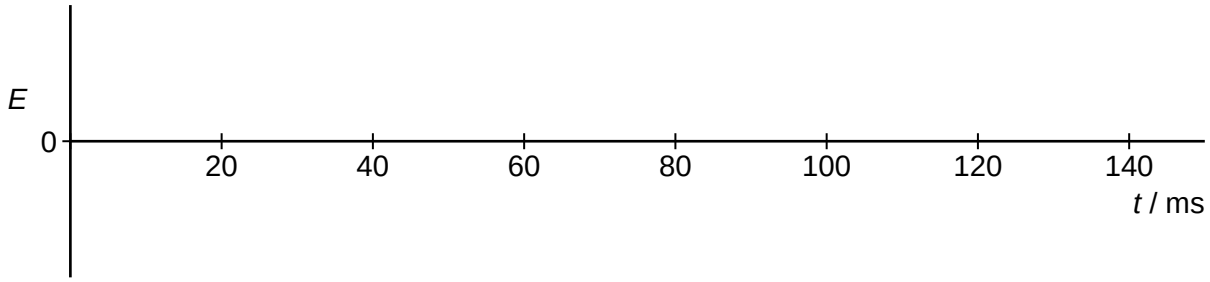


Fig. 7.3

[3]

(ii) Use Lenz's law to explain the direction of the force exerted on the magnet by the current in the coil as the magnet **leaves** the coil.

.....

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..... [2]

(c) Fig. 7.4 shows a thin slice of a conductor carrying a current I in a uniform magnetic field directed into the page. The Hall voltage is measured between points P and Q. The charge carriers in the slice are electrons.

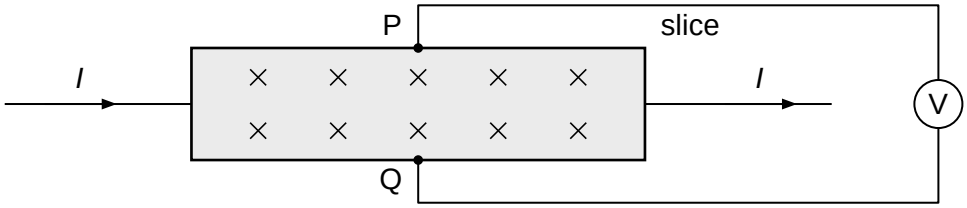


Fig. 7.4

The slice is replaced by a slice with the same dimensions made of a material in which the charge carriers are positive. The number density of the charge carriers in the new slice is one quarter of that in the original slice. The current and the magnetic field are unchanged. State and explain **two** differences between the new Hall voltage and the original Hall voltage.

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..... [3]

[Total: 10]

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8 **(a)** State what is meant by the root-mean-square (r.m.s.) value of an alternating current.

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..... [2]

(b) Four ideal diodes D1, D2, D3 and D4 are used in the bridge rectifier shown in Fig. 8.1.

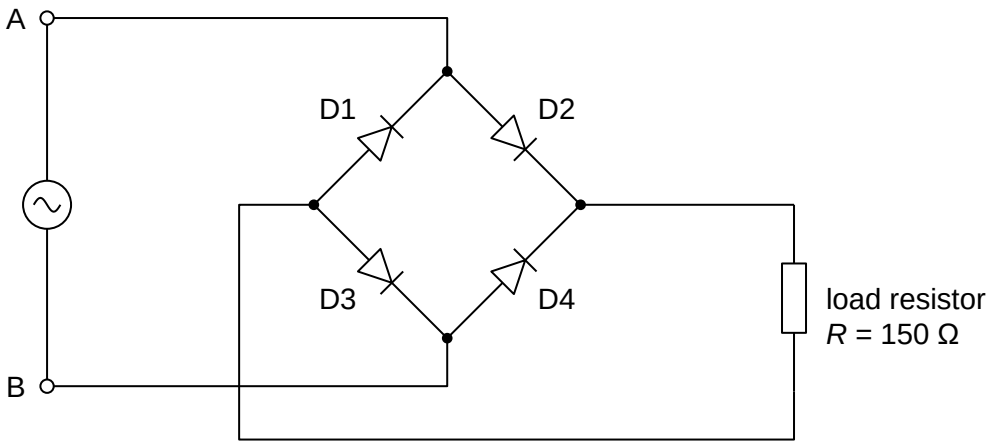


Fig. 8.1

The sinusoidal alternating supply has a peak voltage of 17.0 V and a frequency of 50 Hz. Terminal A of the supply is positive with respect to terminal B between time $t = 0$ and $t = 10$ ms.

Fig. 8.2 shows, as a dashed line, the variation with t of the potential difference (p.d.) V across the load resistor.

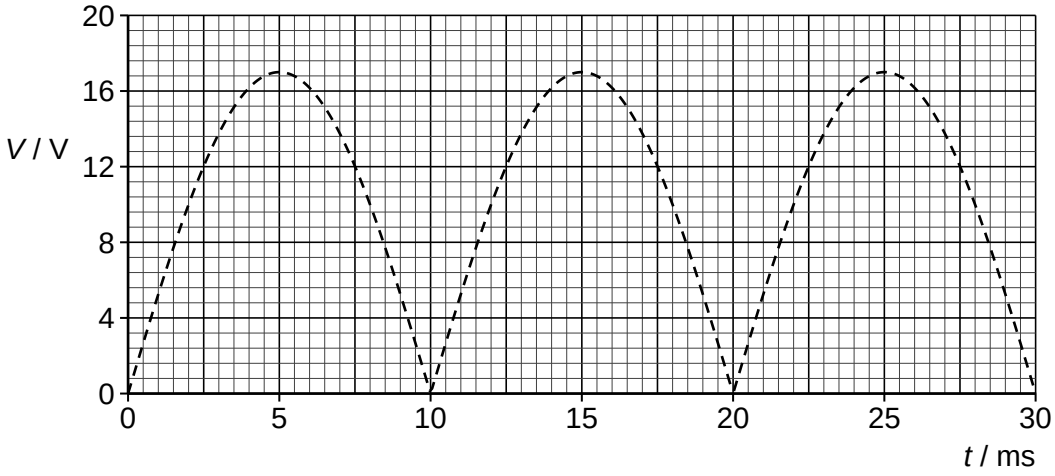


Fig. 8.2

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(i) State the **two** diodes that conduct between $t = 0$ and $t = 10$ ms.

..... [1]

(ii) Calculate the mean power dissipated in the load resistor.

mean power = ... W [2]

(c) Diode D4 develops a fault so that it no longer conducts in either direction.

(i) On Fig. 8.2, draw a solid line to show the variation with t of V between $t = 0$ and $t = 30$ ms. [1]

(ii) Determine the r.m.s. value of V.

r.m.s. value = ... V [2]

(d) A capacitor is now connected in parallel with the load resistor to smooth the output p.d. Diode D4 is still faulty.

Suggest and explain how the variation (ripple) of the smoothed output p.d. compares with the ripple when all four diodes are working.

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..... [2]

[Total: 10]

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9 **(a)** Electrons are accelerated from rest through a potential difference of 4.0 kV. They are then incident on a thin film of graphite in a vacuum, as shown in Fig. 9.1. A pattern of concentric rings is seen on a fluorescent screen.

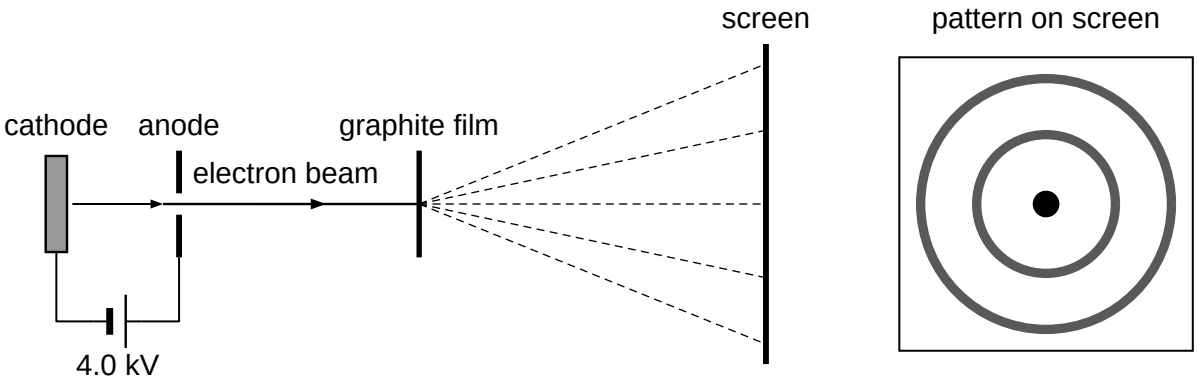


Fig. 9.1 (not to scale)

(i) Calculate the de Broglie wavelength of the electrons.

wavelength = ... m [2]

(ii) The accelerating potential difference is increased to 16 kV. A student predicts that the radius of each ring will decrease to one quarter of its original value. Explain why the student is wrong.

.....

.....

..... [2]

(b) Fig. 9.2 shows four energy levels of an isolated atom.

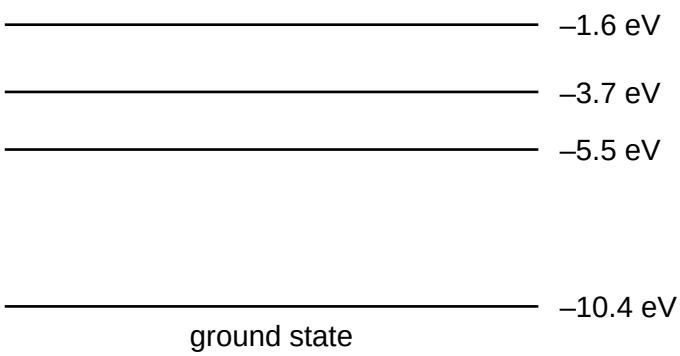


Fig. 9.2

Atoms of a gas are excited so that some of their electrons are in the –1.6 eV energy level.

(i) State the number of different wavelengths of photon that may be emitted as these electrons return to the –10.4 eV level.

number = ... [1]

(ii) Determine the longest wavelength of the emitted photons.

wavelength = ... m [2]

(c) A nucleus of uranium-235 absorbs a slow-moving neutron and undergoes fission according to the equation

$$^{235}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{141}_{56}\text{Ba} + ^{92}_{36}\text{Kr} + 3^{1}_{0}\text{n}.$$

Table 9.1 shows the binding energy per nucleon of each of the nuclei.

Table 9.1

nucleus	binding energy per nucleon / MeV
uranium-235	7.59
barium-141	8.33
krypton-92	8.51

(i) Determine the energy, in MeV, released in this fission.

energy = ... MeV [2]

(ii) Explain, with reference to binding energy per nucleon, why the fusion of two light nuclei can also release energy.

.....

..... [1]

[Total: 10]

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10 **(a)** In positron emission tomography (PET scanning), a tracer containing a β^+ emitter is introduced into the body of a patient. Each emitted positron annihilates with an electron in the tissue, producing two gamma-ray photons.

(i) Explain why the two photons travel in opposite directions.

.....

.....

..... [2]

(ii) The two photons from one annihilation event are detected by detectors D_1 and D_2 in a ring of detectors around the patient, as shown in Fig. 10.1. Point M is the midpoint of the line joining D_1 and D_2.

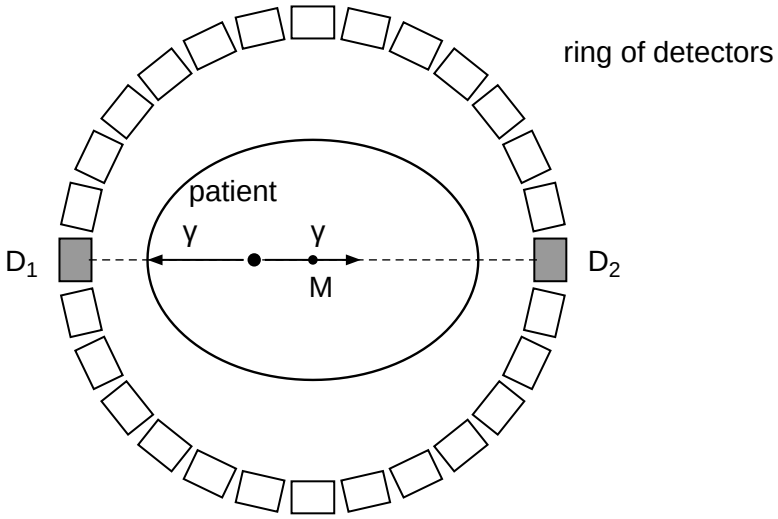


Fig. 10.1

One photon arrives at D_1 0.40 ns before the other photon arrives at D_2.

Determine the distance of the point of annihilation from M.

distance = ... m [2]

(b) Light from a distant galaxy is observed from the Earth. Fig. 10.2 shows the position of one line in the spectrum of hydrogen measured in a laboratory and in the light received from the galaxy.

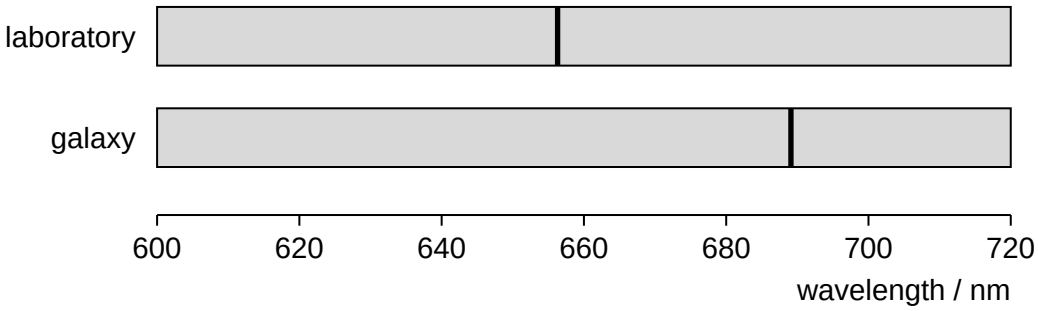


Fig. 10.2

The wavelength of the line is 656.3 nm in the laboratory and 689.1 nm in the light from the galaxy.

(i) Calculate the speed of the galaxy relative to the Earth.

speed = ... m s^{-1} [2]

(ii) The Hubble constant is 2.3×10^{-18} s^{-1}.

Calculate the distance of the galaxy from the Earth.

distance = ... m [1]

(iii) Assume that the galaxy has moved away from the Earth at a constant speed since the Big Bang.

Use your answers in (b)(i) and (b)(ii) to estimate the age of the Universe, in years.

age = ... years [2]

[Total: 9]

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